

6 Beyond linear regression Non-additivity, non-linearity and mediation

6.1 Introduction

Sometimes the complexities of life entail that the linear regression model paints a *too* simplistic picture of reality. If under these circumstances we hold on to the linear model, the estimate of b_1 for x_1 on y could be biased. The purpose of this chapter is to illustrate the ways in which we may adapt the linear regression model to these more complicated circumstances. We first address the violations of assumption B1 (section 6.2) and B2 (sections 6.3 to 6.5). Finally, section 6.6 takes up the issue of mediation which will be explained when we get there.

6.2 Non-additivity (assumption B1): Interaction/moderator regression models

In all the $y = b_0 + b_1x_1 + b_2x_2$ regressions in this book so far, it has been tacitly assumed that the various regression coefficients, or effects, are of similar size for every subgroup of observations. Indeed, this is what the additivity assumption (B1) means. And without a theory or common sense telling us anything to the contrary, this is a sound point of departure for statistical analysis. In some cases, however, we have reason to suspect that this assumption of “constant bs” does not sit well with reality. Moreover, we may even expect that the size of b_1 is larger or smaller for subgroup A than it is for subgroup B in real life. Below we learn how to examine if this is the case and how to deal with it.

To set the stage, consider a regression between earnings (y) and years of education (x_1) and gender (x_2) for a random sample of 300 employees. (We omit other x -variables for simplicity, but without loss of generality.) Let's say b_1 is 1 000, meaning that one more year of education entails 1 000 more in earnings given a dynamic interpretation, controlling for gender. So far, so good. The additivity assumption – i.e. the constant bs – now says that years of education's effect on earnings is the *same* for men and women. But is it necessarily so in real life? The short answer is no. The longer answer is that research has shown that women, as compared with men, benefit more in terms of earnings from investing in one more year of education. Graphically we may see the pattern in Figure 6.1. Note that although women earn less than men on average (the solid line is lower on the y -axis), the relationship between education and earnings is steeper (i.e. has a larger b_1) for women. That is, the additivity assumption (B1) is violated. Additivity thus means *parallel* regression lines.

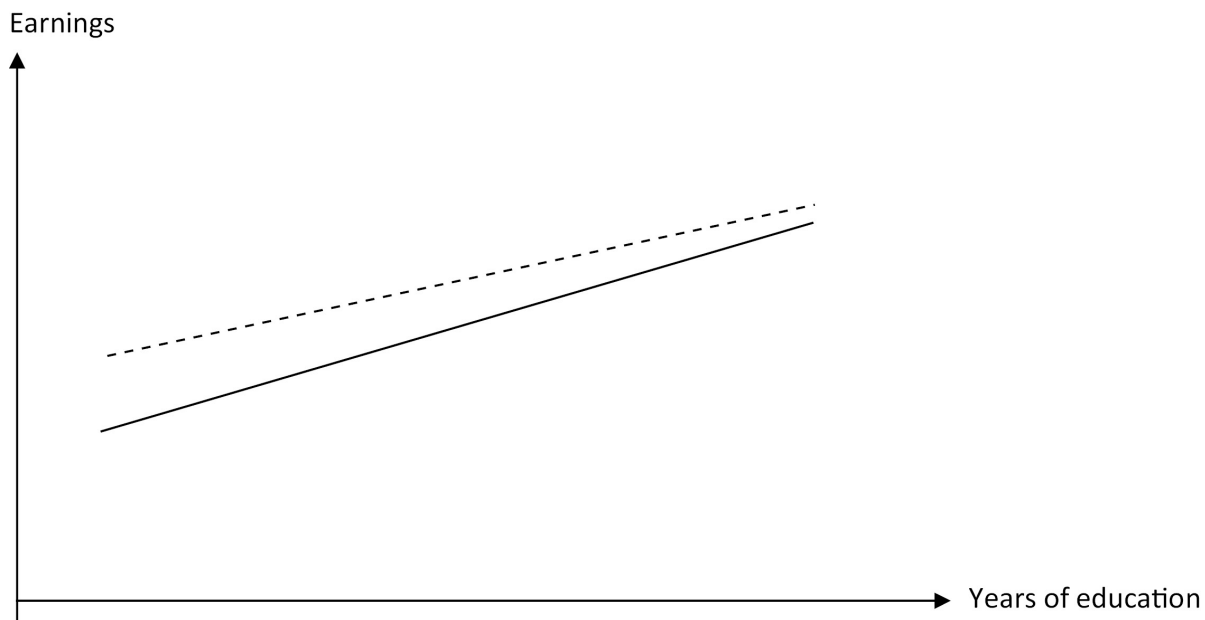


Figure 6.1 Stylized relationship between earnings (y) and years of education (x_1) for men (dashed line) and women (solid line).

How do we unearth relationships of the kind depicted in Figure 6.1? An obvious solution is to do two separate regressions: one for women and one for men. This procedure has one drawback, however, as it requires a rather large number of observations. The more common procedure is to estimate an interaction or moderator regression model. (Interaction and moderator regression models are terms used interchangeably in the literature.) The procedure involves first creating, and then adding, an *interaction variable* to the regression model. Here's an example on how to do it.

We use a new dataset (`stud_phys_17`): a random sample of students who have answered a questionnaire. The y is weekly hours of exercise (`phys_act_h`), and the x -variables are gender (women = 1, men = 0) and the students' friends' exercise levels (`friend_phys`; a 0 to 4 scale where 0 denotes inactive friends and 4 very active friends).¹ We hypothesize that men and students with physically active friends spend more time exercising themselves than do women and students with inactive friends. [Stata-output 6.1](#) presents the results for the standard, additive model.

The additive model. The output says that men exercise for 1.4 *more* hours per week than women *ceteris paribus* (remember the coding of gender), and similarly that a one-unit increase in friends' exercise levels entail 0.6 hours more exercising per week in a dynamic sense. Both coefficients are statistically significant at $p < 0.0001$, suggesting that both alternative hypotheses (i.e. one for gender, and one for friends' activity levels) are indirectly supported.

This additive regression tacitly assumes that the regression coefficient for friends' exercise levels (0.593) is of the same size for men *and* women. But let's suppose some researcher has published a paper showing that these coefficients are of different sizes. To examine this, we use a mechanistic procedure available in every statistics program. We first create an interaction term (i.e. a product variable) between the two variables of interest. We then add this interaction term to our initial regression model in [Stata-output 6.1](#). [Stata-output 6.2](#) illustrates. Before the regression command, Stata is told to generate the variable `wom_f_p`, i.e. the product of gender and `friend_phys`. (In Stata commands `*` = $\times$.) This product term variable is then added to the regression command in the usual way.²

The interaction model. The key to interpreting interaction models with a dummy is to recall how the dummy is coded, i.e. 1 = women and 0 = men in this case. The point is that the regression coefficient of interest (*friend_phys*) refers to the subgroup with the value of 0 on the dummy. That is, for men the effect of friends' exercise levels on hours of exercise is 1.09. For women, in contrast, the analogous effect is 0.30, i.e. the effect of *friend_phys* (1.09) plus the effect of the product term *wom_f_p* (-0.79 ; $1.09 - 0.79 = 0.30$).³ Conclusion: The effect of friends' exercise levels on students' hours of exercise is *larger* for men than for women. The coefficient for gender refers to students with 0 on the friends' exercise levels variable. The difference between the genders in exercise hours for this group is about half an hour (0.58) in women's favor, but this difference is not significant at conventional levels.

Interaction effects are more easily understood and interpreted when presented in graphs. Figure 6.2 is based on the regression in Stata-output 6.2.⁴ Note the steeper regression line for men, suggestive of a larger regression coefficient for this subgroup. The figure also shows, on the left-hand side, the small gender difference for low values on the friends' exercise levels variable.

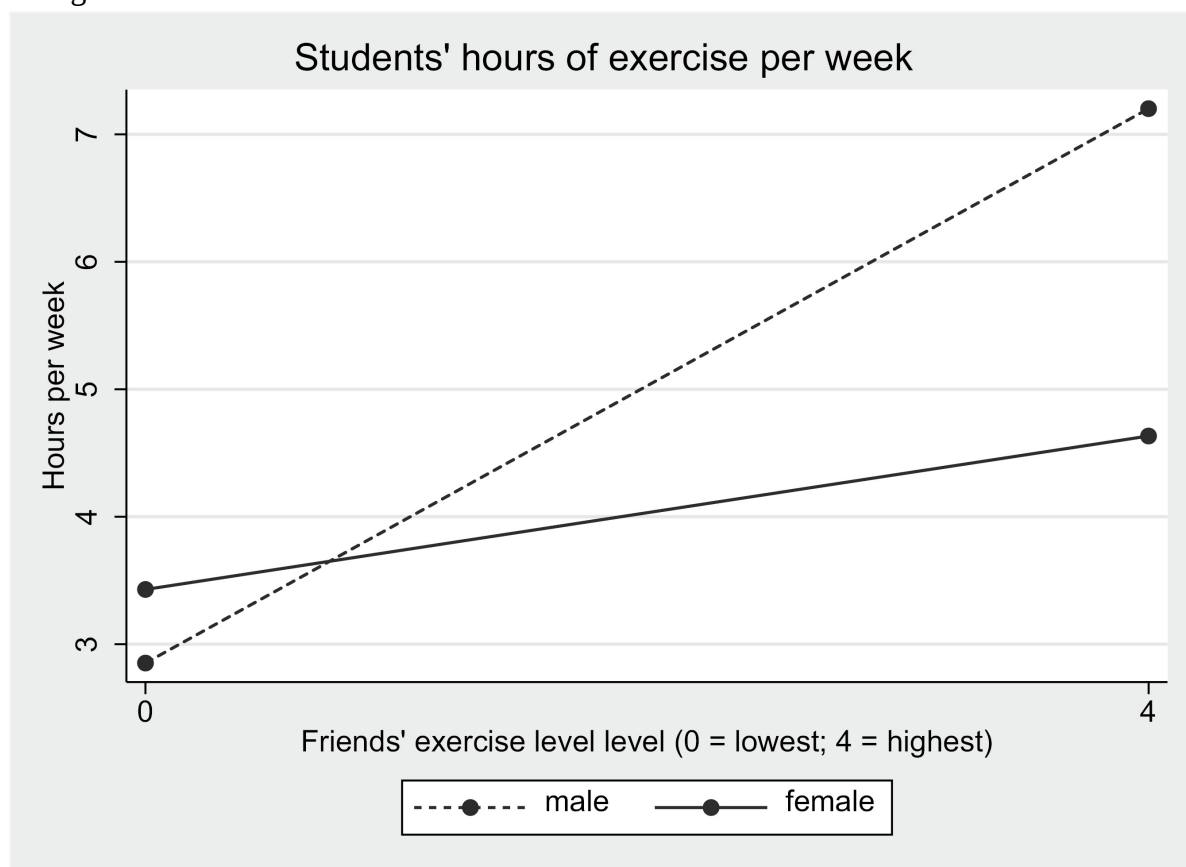


Figure 6.2 Visual illustration of the interaction effect between gender and friends' exercise levels, based on the regression model reported in Stata-output 6.2.

One issue remains. Why did we conclude that there indeed was an interaction effect to speak of? The answer is that the regression coefficient for the product term (*wom_f_p*) had a *significant* effect on hours of exercise. This is also the general idea. When an interaction variable (or product term) has a significant effect, we conclude that the simpler additive model is inadequate and gain support for the non-additive model (i.e. interaction model). In contrast, when a product term has an insignificant effect, we keep the simpler, additive model.⁵ This is Occam's razor or the principle of parsimony in practice: We prefer simpler models to more complicated ones when we have no reason to suspect that the simpler models are erroneous. In this case we had such reason.

Let's take stock: Sometimes reality bites in the sense that a regression coefficient in real life is of varying size for different subgroups of observations. If we then force our regression to be additive (i.e. commit a so-called specification error), we get a mismatch between reality and the regression, causing our regression coefficient to be biased. To rectify this, we use an interaction/moderator regression model that allows a regression coefficient to be of *unequal size* for different subgroups in the data. When do we test for an interaction effect? When there is a theoretical or common-sense reason to suspect that an additive regression model is a too simple representation of reality.⁶

6.3 Non-linearity 1 (assumption B2): Quadratic regression models

Non-linearity concerns *how* the independent variable of interest, x_1 , is related to y .⁷ The problem is that this relationship is not linear, as in the stylized relationship between x_1 and y in Figure 6.3. (We now assume for simplicity, as we also do in sections 6.4 and 6.5, that our regression is additive, i.e. that no interaction terms are necessary.) How do we adapt the regression model to the non-linear scenario in Figure 6.3?

Thankfully there is also a very straightforward and mechanistic solution to this problem. The strategy involves adding a quadratic or squared term to the regression. (The square of x is x multiplied with x , i.e. x^2 .) By doing this, we create an up-and-then-down regression line like the one in Figure 6.3.

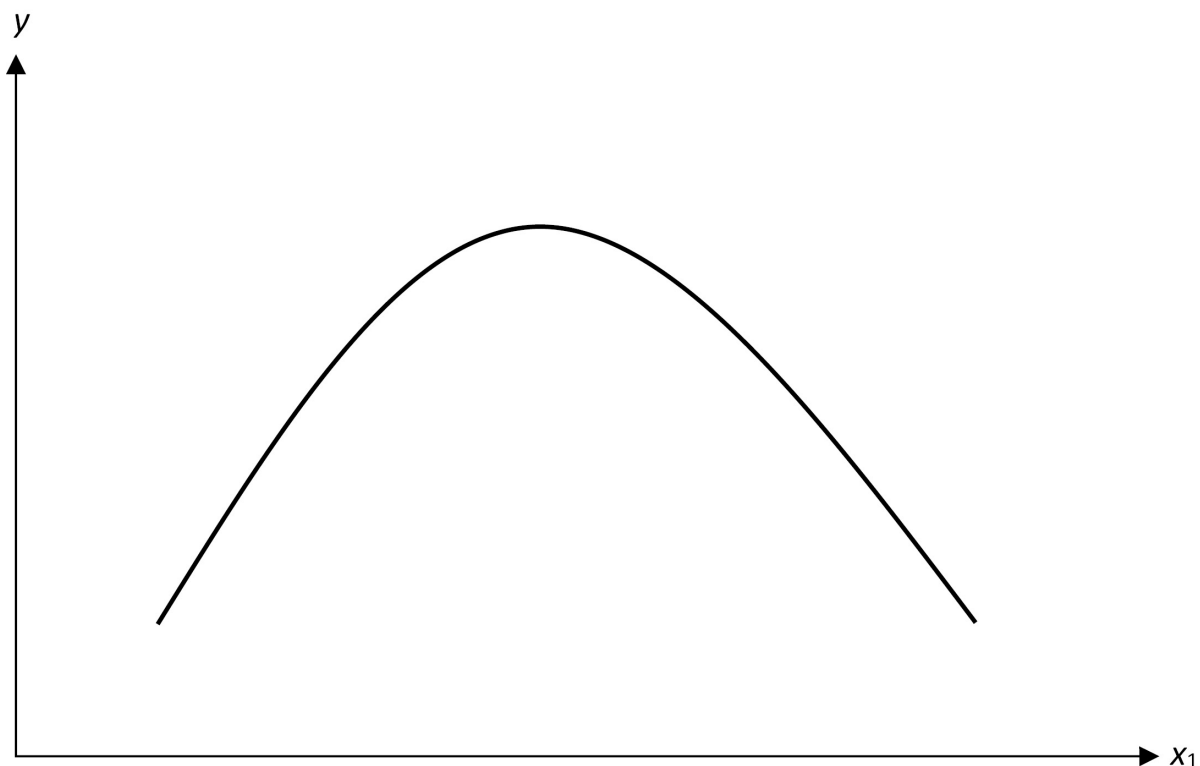


Figure 6.3 A non-linear relationship (i.e. an inverse U-relationship) between x_1 and y .

Below is an example using the soccer player data. The y is again yearly earnings, and the independent variable is age. (More independent variables do not cause additional difficulties.) [Stata-output 6.3](#) shows the computation and results. The quadratic term is first created and then added to the regression in the usual manner. The key to interpretation lies in the sign of the b s. If the original variable (age) has a positive b and its quadratic (agesquared) a negative b , we have the pattern depicted in Figure 6.3. This implies an inverse U-relationship between age and earnings: Players earn more as they age in the beginning of their careers, but at some point their earnings reach a top point

and then start to decline with aging. This maximum point is calculated by the formula $-b_1/(2 \times b_2)$, where b_1 here refers to age and b_2 refers to agesquared. We thus have $-45\,292/(2 \times -726) = 31.19$; the players reach their top earnings potential at 31 years of age.

Let's take a moment. Why include a quadratic (squared term) in a regression in the first place? The answer is that we based on theory, prior research or common sense suspect that the relationship between x_1 and y has an inverse U-form or U-form in real life. (Prior research on the determinants of soccer players' earnings does suggest this inverse U-relationship.) The evidence of such a non-linear relationship is, in short, a significant regression coefficient for the quadratic term. We thus keep a *significant* quadratic term in the regression, since this suggests a non-linear relationship between x_1 and y in real life. If the quadratic term is insignificant, in contrast, we have no case for non-linearity and again prefer the simpler, linear model using Occam's razor. In our case, the b for agesquared is significant at $p = 0.001$, implying a non-linear relationship. Preferring a linear model over a non-linear one when there is evidence of non-linearity amounts to the same kind of specification error as holding on to an additive model when there is evidence of non-additivity (see [section 6.2](#)). [Figure 6.4](#) contrasts the linear regression (solid regression line) with the quadratic regression (dashed regression line).⁸

When the effect of x is positive and the effect of x^2 is negative, the relationship between x and y has an inverse U-form. When the effect of x is negative and the effect of x^2 is positive, the relationship between x and y has the U-form. In the latter case the formula above yields a minimum point.⁹

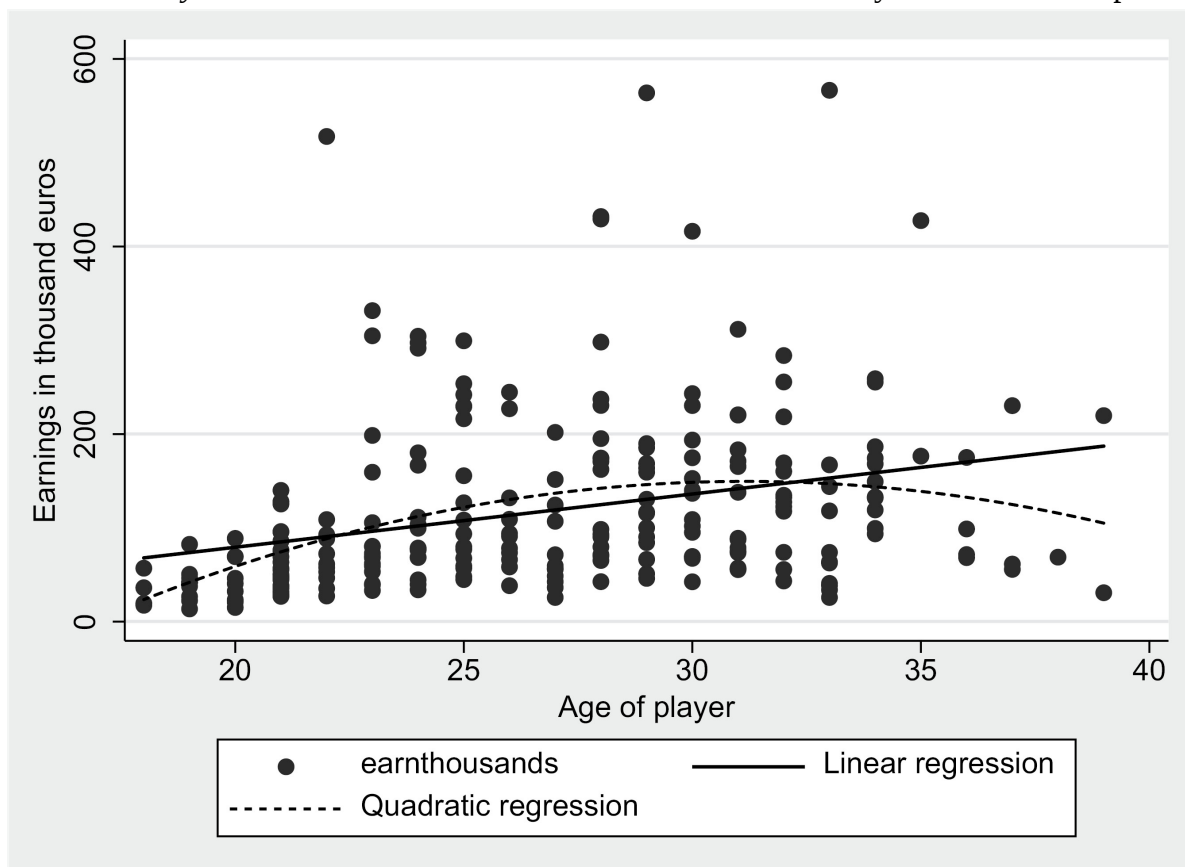


Figure 6.4 Comparison of linear and quadratic (i.e. non-linear) regression between age and earnings. 242 players in the Norwegian version of PL. Graph generated in Stata.

6.4 Non-linearity 2 (assumption B2): Sets of dummy variables

Non-linearity between x_1 and y can be addressed in various ways, of which [section 6.3](#) described one. Another approach is the use of sets of dummy variables. This approach was actually shown in [section 3.4](#), and we repeat and adjust it here using the newly analyzed soccer player data. Again, the variables of interest are age and earnings. The idea is to create a dummy for each *age group* and then compare these age groups' earnings to a reference age group. (The age distribution of the players ranges from 18 to 39 years; not shown.)

The regression appears in [Stata-output 6.4](#). Before doing the regression, I first created the five age group dummies age 21 to 23, age 24 to 26, age 27 to 29, age 30 to 32 and age 33 or more.¹⁰ I did not create the dummy for age 18 to 20, which thus serves as the reference group or -category.

The coefficient for age21_23 says that players aged from 21 to 23 years earn about 46 000 € more on average than players aged between 18 and 22 (the reference). The latter reference group earns about 41 000 € on a yearly basis (the constant). The remaining dummy coefficients follow the same pattern in terms of interpretation. For example, players aged from 30 to 32 years earn 102 000 € more than the reference category on average. The non-linearity in the earnings-age association is present in that the three oldest age groups (27–29, 30–32 and 33 or more) have rather equal annual earnings. That is, the earnings do not increase steadily with aging as implied by linearity.

In summary, both the quadratic and the sets of dummy variables approach may be used to handle non-linearity between x_1 and y . The latter procedure in general requires more observations, though. A third approach for addressing non-linearity, coming up now, is the use of logarithms.

6.5 Non-linearity 3 (assumption B2): Use of logarithms

Let's say we want to find out how the price of TVs is related to their size in inches. Given access to data containing these variables, it is tempting to regress price on inches in the usual manner shown so far in this book. And this is exactly what is done in [Stata-output 6.5](#) for 121 TVs in the Norwegian market. (The name of the data file is tv_price.)

Since we are comparing different TVs, the regression coefficient suggests that a 51-inch TV costs 527 NOK more on average than a 50-inch TV. The independent variable explains 69% of the variation (variance) in TV prices. But is this relationship in fact linear? The plot in [Figure 6.5](#) examines this using the scatterplot smoother introduced in [section 5.3](#).

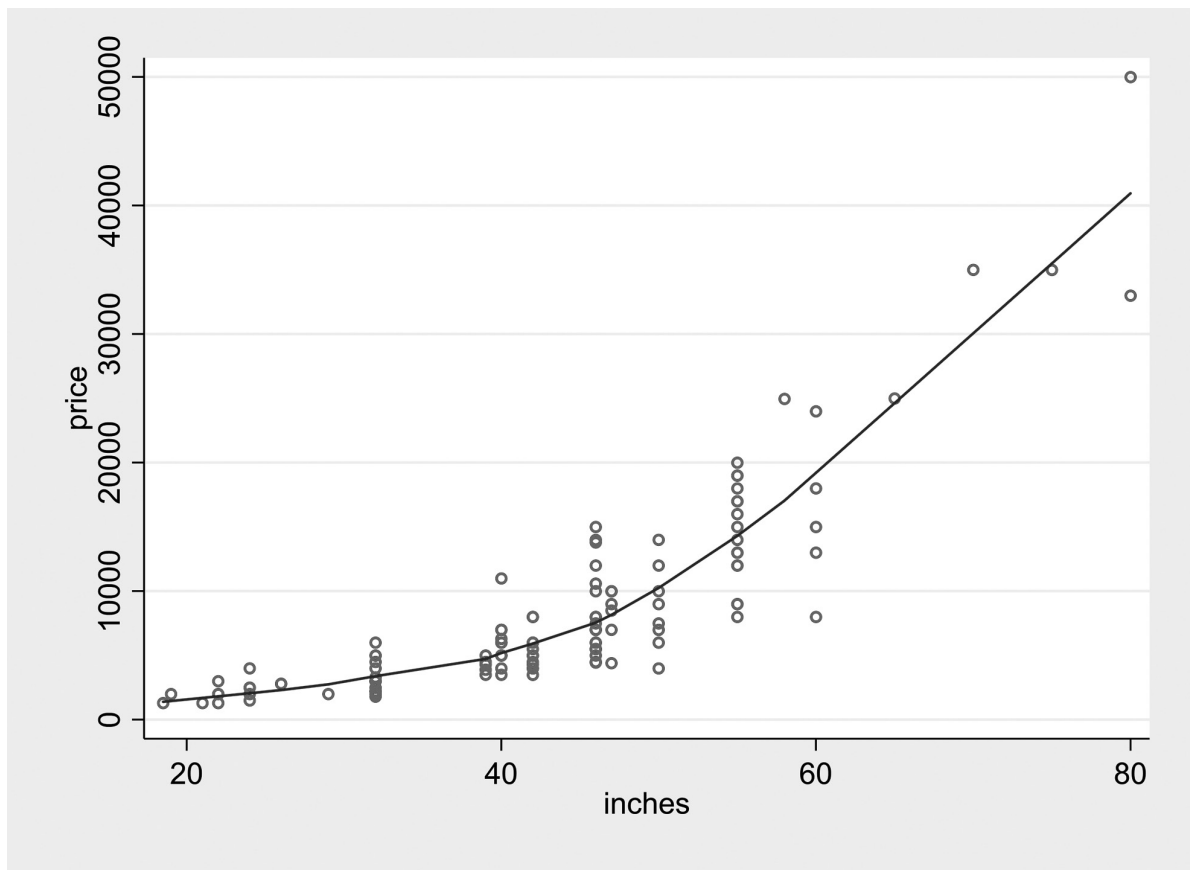


Figure 6.5 TV price by TV size in inches, with scatterplot smoother. 121 TVs in the Norwegian market. Graph generated in Stata.

The answer is no. The relationship between the two variables is non-linear. But since the regression line increases in an accelerating way and has its minimum at the lowest x -value (i.e. $x = 18.5$), the quadratic regression is not optimal. In contrast, the (sets of) dummy variable approach might be used if there are enough observations in the data. However, there is a more elegant way to proceed: to use logarithms.

Using logarithms in regression analysis means using a transformation of y or x_1 (or both) instead of the original variables.¹¹ Taking the natural logarithm of a number is to crunch it through a formula to become a new number. In this way, the logarithm of 1 becomes zero and the logarithm of 5 becomes 1.609. Similarly, the logarithm of 100 is 4.605 and the logarithm of 10 000 is 9.210. In other words, when large numbers are expressed on the logarithmic scale they are much closer in magnitude to small numbers than when stated on their original scale.¹² This trite fact has some nice consequences for regression, especially regarding how to solve non-linearity. (Using logarithms goes by many names, such as “logging a variable”, “using a log” or simply “logging”. And when using the logarithm of y in a regression, we generally write $\log y$ or $\ln y$.) [Stata-output 6.6](#) shows the regression between the log of TV prices and TV size in inches. Note that the logging procedure must precede the regression command (“gen log_price = ln(price)”).

The only thing of note in the output is that the regression coefficient has changed in magnitude. It is now ≈ 0.06 and implies that a 51-inch TV costs 6% more on average than a 50-inch TV. This is also the general idea: When y is expressed in logs, the regression coefficient refers to change in y in *percent* by a one-unit increase in x , i.e. b has a *relative* interpretation.

Back to the problem. Does this solve the non-linearity in [Figure 6.5](#)? [Figure 6.6](#) shows the analogous scatterplot smoother for the log of TV price and TV size. It shows a fairly linear

relationship between the two variables. Logging y in regressions like this thus often amounts to “linearizing” a non-linear relationship between x_1 and y .¹³ When should we use the log of y instead of y in a regression? As before, theory, prior research or common sense guide us on the real-life functional relationship between x_1 and y . When this on occasion provides us with few pointers, however, checking the matter empirically – as done above – should be a straightforward (and mandatory?) enterprise. Generally, logging y is preferable if the real-life functional relationship between x_1 and y is of one of the forms depicted in [Figure 6.7](#).

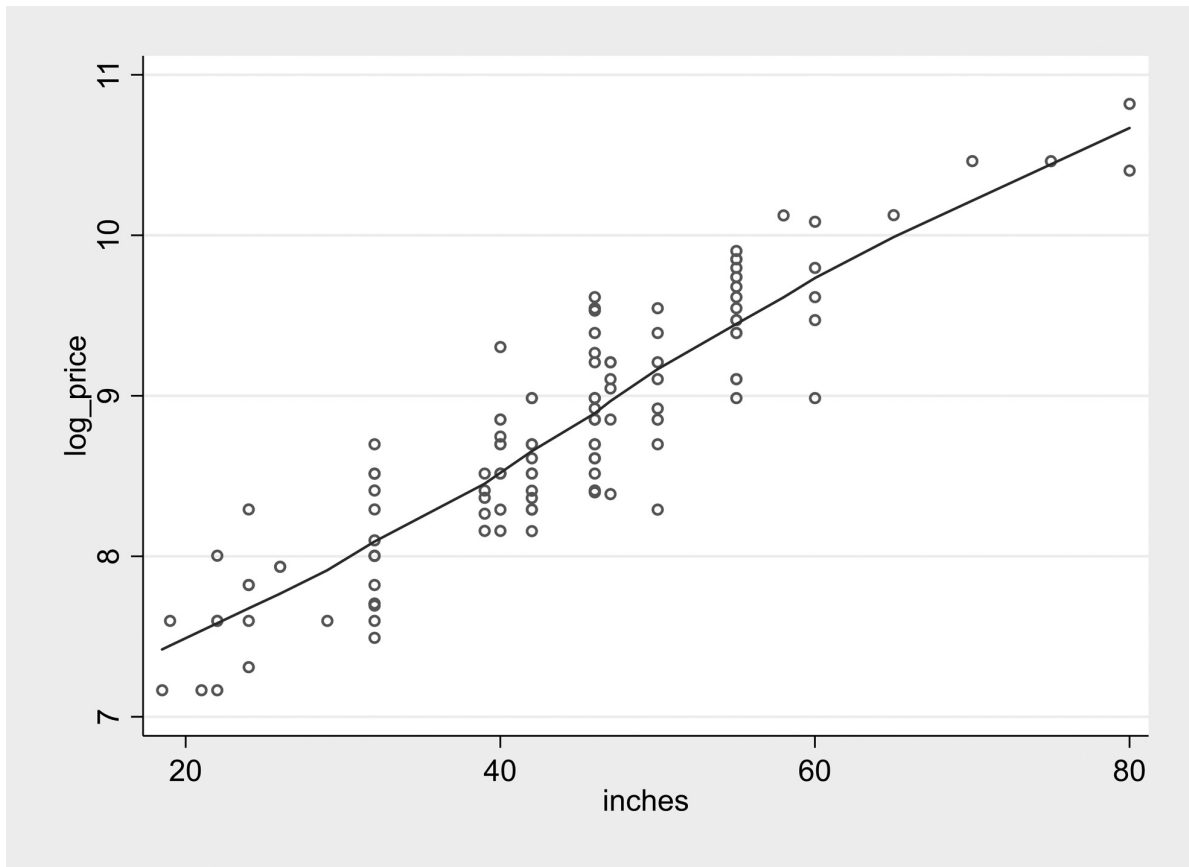


Figure 6.6 TV price (logged) by TV size in inches, with scatterplot smoother. 121 TVs in the Norwegian market. Graph generated in Stata.

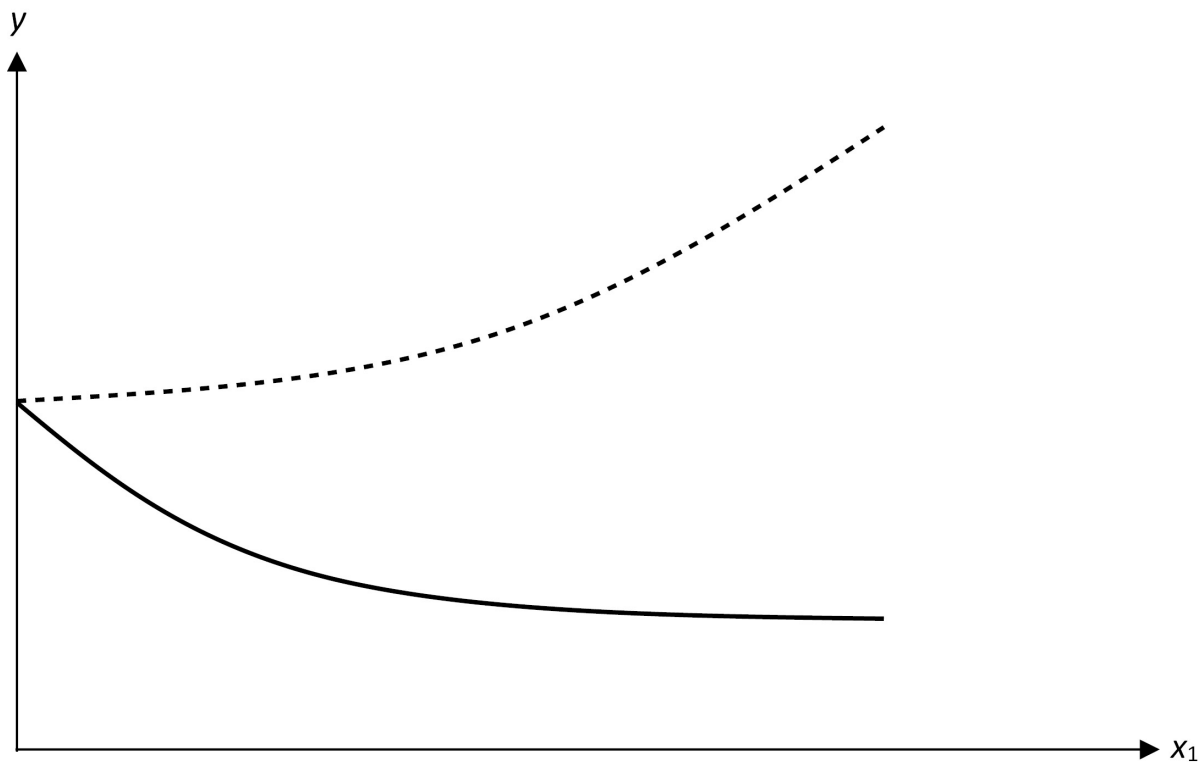


Figure 6.7 Dashed line implies $b_1 > 0$ if y is logged, and solid line implies $b_1 < 0$ if y is logged.

The use of dummies when y is logged. In [Stata-output 6.7](#) the variable *smart* (short for smart TV or not; yes = 1, no = 0) is added to the regression in [Stata-output 6.6](#). Using the interpretation above we are tempted to say that smart TVs cost 49% more than non-smart TVs *ceteris paribus*. However, this interpretation only holds when b is smaller than roughly ± 0.20 . For a larger b , we need a formula to get to the correct percentage interpretation (see Halvorsen and Palmquist, 1980), namely

$$100 \times (e^b - 1).$$

This yields $100 \times (2.71828^{0.494} - 1) = 63.9\%$ in our case. Smart TVs cost 64% more than non-smart TVs, controlling for TV size. ($e^{0.494}$ equals taking the antilogarithm of 0.494.)

The last two regressions are called semi-logarithmic regressions or semi-elasticity models. Their “cousins”, where x_1 (and x_2 and so on) and y are logged, are called double-logarithmic models. The only thing changing for the latter models is the interpretation of the regression coefficient, which is change in y in *percent* given a *one-percent* increase in x .¹⁴ The last thing to note before proceeding with logarithmic regression modeling is that logging is only possible for strictly positive values. That is, the logarithm of 0 is not defined.¹⁵

6.6 Mediation models

Except for in [section 6.2](#), we have so far estimated multiple regressions of the kind depicted in [Figure 6.8](#). Here, the effects of x_1 and x_2 on y , as expressed by b_1 and b_2 (i.e. the arrows pointing towards y), are controlled for each other and share a correlation (i.e. the bidirectional arrow). That is, the effect of x_1 , which often is of main interest, may be thought of as x_1 ’s unique effect on y . And similarly for x_2 . But what if the real-life association between the three variables looks more like the stylized model depicted in [Figure 6.9](#)?

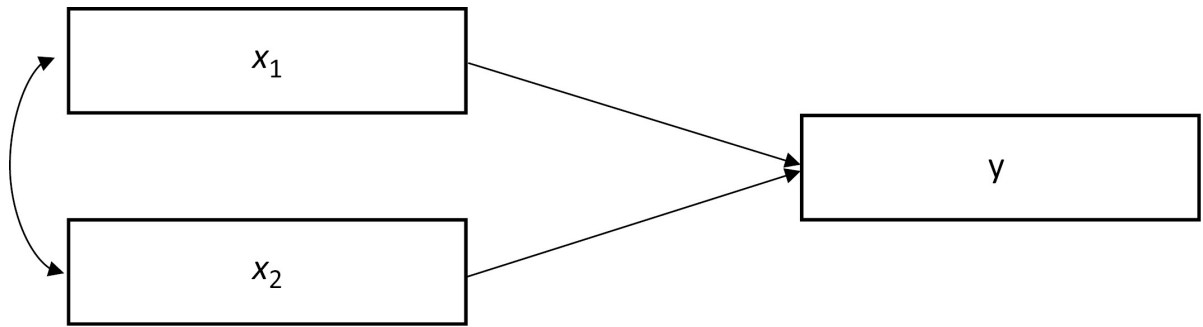


Figure 6.8 Traditional multiple regression model where the effects of the two independent variables are controlled for each other. The bidirectional arrow symbolizes the fact that the two independent variables share a correlation.

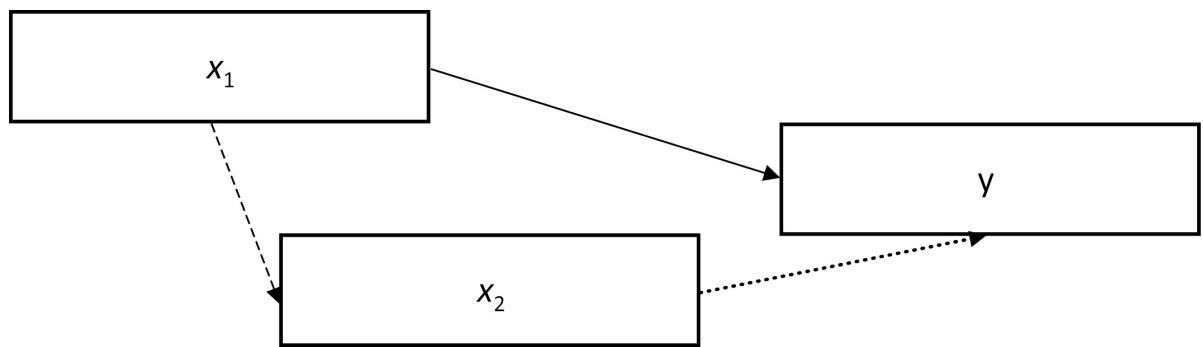


Figure 6.9 Path model where x_1 affects y directly (solid arrow) and indirectly through x_2 (dashed arrow and then dotted arrow).

Figure 6.9 suggests that x_1 has a *direct* effect on y (solid arrow) as well as an *indirect* effect on y via x_2 (dashed and dotted arrow). If we were to estimate this model by the usual multiple regression, however, the direct effect would be the only effect picked up by the regression. That is, since the regression would *not* pick up the indirect effect, it would *underestimate* the total effect of x_1 on y . (Total effect = direct effect + indirect effect.)

To examine (path) models like in Figure 6.9, we need to estimate a *set* of different regression models. The following example illustrates for the simplest type of path model, i.e. the model in Figure 6.9. Variable x_1 is parents’ interest in exercising (par_exe_int), x_2 is students’ interest in exercising (exercise_int) and y is students’ hours of exercise per week (phys_act_h).¹⁶ The data are fem_stud_sport.dta. The two independent variables are *indexes* ranging from 0 (minimum interest) to 6 (maximum interest). (See section 2.5 if you have forgotten about indexes.) The path model is presented in Figure 6.10.

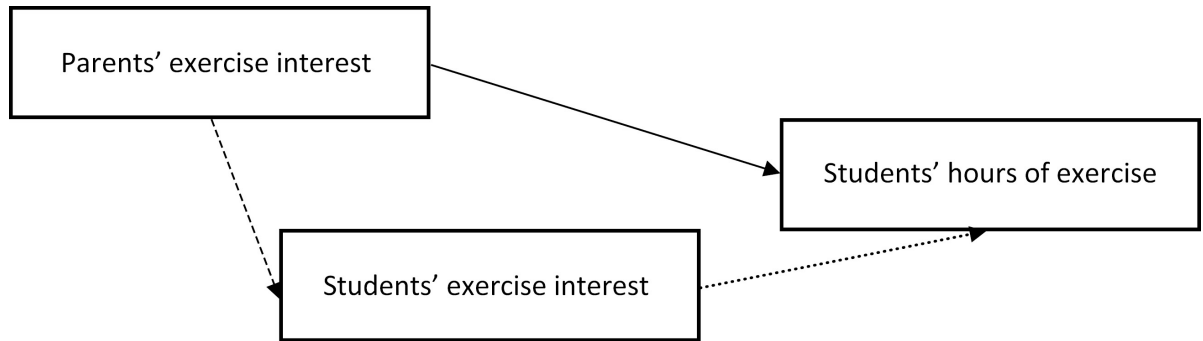


Figure 6.10 Path model where parents’ exercise interest affects students’ hours of exercise directly (solid arrow) and indirectly through students’ exercise interest (dashed and then dotted arrow).

Three regressions must be carried out to capture the (a) total effect, (b) direct effect and (c) indirect effect of parents' exercise interest in [Figure 6.10](#):

- (1) Students' hours of exercise on parents' exercise interest *only*
- (2) Students' hours of exercise on parents' exercise interest *and* students' exercise interest
- (3) Students' exercise interest on parents' exercise interest (dashed arrow)

Regression (1) estimates the total effect, and regression (2) estimates the direct effect. The indirect effect (c) is found by multiplying the regression coefficient for students' exercise in (2) (dotted arrow) by the regression coefficient in (3) (dashed arrow). The results of the regressions appear in [Stata-output 6.8](#).

The top regression (1) suggests that the total effect is 0.465. The middle regression (2) suggests a direct effect of 0.201. Finally, the indirect effect is 0.264 (1.327×0.199) based on the middle (2) and bottom (3) regression. (Note again that total effect = direct effect + indirect effect.)

The upshot? Were we to examine the effect of parents' exercise interest on students' hours of exercise by standard multiple regression, we would suggest an effect of 0.201 (i.e. the direct effect). That is, we would underestimate the total effect of this variable, which is 0.465 (i.e. direct + indirect effect).¹⁷

In many applications the x_2 variable – here: students' exercise interest – is called a *mediator*. Another term for it is a *mechanism*, i.e. a variable that (at least partially) explains the statistical relationship between x_1 and y . Here, students' exercise interest (x_2) explains 57% of the association between parents' exercise interest (x_1) and students' hours of exercise, y ($((0.4652 - 0.2007)/0.4652 \approx 0.569)$). In this sense, x_2 is a *partial* mediator.

When do we employ a path (mediator) model instead of the usual multiple regression model? The answer is when there is a sounder theoretical justification for the former in terms of what happens in real life. The purpose of this section has been to provide the reader with a first take on path models using a simple example. However, the principle shown holds for models including several controls as well as several potential mediators. Of course, in these expanded models, the mathematical and statistical difficulties increase, but to accommodate those we have statistics programs at our disposal.

In certain textbooks, especially those going by the name of econometrics, the x_2 or mediator is called a “bad control variable”.¹⁸ A bad control is a variable that lies between the independent variable of interest, x_1 , and y in the causal chain or process going from left to right in the above figures. That is, including a bad control in a regression will *underestimate* the total effect of x_1 on y . For this reason, the advice goes, bad controls must be discarded from the regression. Another and perhaps more constructive approach would be to use a path (mediator) model instead.

6.7 Summary and further reading

[Chapter 6](#) first dealt with the regression assumption of additivity, which says that the coefficients in a multiple regression are of the same size for every subgroup of observations. When this assumption does not sit well with reality, we instead opt for a regression that includes one or more interaction terms. (This is often called a moderator or multiplicative regression model.) When this interaction term's regression coefficient is statistically significant (suggesting a violation of additivity), we prefer the interaction regression model. When the interaction term's regression coefficient is insignificant, in contrast, we stick to the simpler, additive regression model based on the principle of parsimony.

The middle of [Chapter 6](#) illustrated three ways in which to handle a non-linear relationship between x_1 and y , i.e. a violation of the linearity assumption in linear regression analysis. The three strategies offered are sometimes equivalent and sometimes not. When there are no signs of non-linear effects, we prefer the simple, linear model in keeping with the parsimony principle.

The final part of [Chapter 6](#) introduced mediator or path models. These models are to be preferred when we have reason to believe that our independent variable of interest, x_1 , also has *indirect* effects on y working through other independent variables, such as x_2 .

Interaction or moderator regression models are covered in most of the books and papers mentioned in the “further reading” sections so far. Mitchell (2012) is a very useful and graphically oriented book on the subject using Stata. Non-linearity is also covered in every textbook. Baron and Kenny (1986) is the standard reference to mediator models, and a recent and very readable review and introduction to the subject is given by Pieters (2017).

```
. use "C:\Users\700245\Desktop\temp_myd\stud_phys_17.dta"
```

```
. reg phys_act_h gender friend_phys
```

Source	SS	df	MS	Number of obs	=	657
Model	456.164947	2	228.082473	F(2, 654)	=	23.58
Residual	6325.32181	654	9.67174589	Prob > F	=	0.0000
				R-squared	=	0.0673
				Adj R-squared	=	0.0644
Total	6781.48676	656	10.3376323	Root MSE	=	3.1099

phys_act_h	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
gender	-1.415949	.2547573	-5.56	0.000	-1.91619	-.9157081
friend_phys	.5927305	.1633656	3.63	0.000	.271946	.9135149
_cons	4.125651	.4679033	8.82	0.000	3.206877	5.044425

ta-output 6.1 Regression of hours of exercise per week (phys_act_h) on gender (gender) and friends' exercise levels (friend_phys).

```
. generate wom_f_p = gender * friend_phys
```

```
. reg phys_act_h gender friend_phys wom_f_p
```

Source	SS	df	MS	Number of obs	=	657
Model	508.47031	3	169.490103	F(3, 653)	=	17.64
Residual	6273.01645	653	9.60645704	Prob > F	=	0.0000
				R-squared	=	0.0750
				Adj R-squared	=	0.0707
Total	6781.48676	656	10.3376323	Root MSE	=	3.0994

phys_act_h	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
gender	.5774496	.8912158	0.65	0.517	-1.172545	2.327444
friend_phys	1.087477	.267327	4.07	0.000	.5625532	1.612402
wom_f_p	-.786476	.3370497	-2.33	0.020	-1.448308	-.124644
_cons	2.851303	.7181326	3.97	0.000	1.441175	4.26143

ta-output 6.2 Regression of hours of exercise per week (phys_act_h) on gender (gender) and friends' exercise levels (friend_phys) and their product variable (wom_f_p).

```
. use "C:\Users\700245\Desktop\temp_myd\football_earnings.dta", clear
. gen agesquared = age * age
. reg earnings_E age agesquared
```

Source	SS	df	MS	Number of obs	=	242
Model	2.7544e+11	2	1.3772e+11	F(2, 239)	=	17.29
Residual	1.9036e+12	239	7.9648e+09	Prob > F	=	0.0000
Total	2.1790e+12	241	9.0416e+09	R-squared	=	0.1264
				Adj R-squared	=	0.1191
				Root MSE	=	89246

earnings_E	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
age	45292.13	12193.37	3.71	0.000	21271.93	69312.33
agesquared	-726.4627	222.5748	-3.26	0.001	-1164.922	-288.0039
_cons	-556343.5	163076.3	-3.41	0.001	-877594	-235093.1

ta-output 6.3 Regression of earnings (earnings_E) on player age (age) and square of player age (agesquared).

```
. reg earnings_E age21_23 age24_26 age27_29 age30_32 age33ormore
```

Source	SS	df	MS	Number of obs	=	242
Model	2.5654e+11	5	5.1308e+10	F(5, 236)	=	6.30
Residual	1.9225e+12	236	8.1461e+09	Prob > F	=	0.0000
Total	2.1790e+12	241	9.0416e+09	R-squared	=	0.1177
				Adj R-squared	=	0.0990
				Root MSE	=	90256

earnings_E	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
age21_23	46350.58	22669.46	2.04	0.042	1690.229	91010.92
age24_26	86192.5	23134.42	3.73	0.000	40616.15	131768.8
age27_29	99667.94	23134.42	4.31	0.000	54091.59	145244.3
age30_32	102007.4	23223.17	4.39	0.000	56256.19	147758.6
age33ormore	99896.96	24367.35	4.10	0.000	51891.65	147902.3
_cons	40859.49	18819.6	2.17	0.031	3783.62	77935.36

ta-output 6.4 Regression of earnings (earnings_E) on age group dummies.

```
. use "C:\Users\700245\Desktop\temp_myd\tv_price.dta", clear
. reg price inches
```

Source	SS	df	MS	Number of obs	=	121
Model	4.9239e+09	1	4.9239e+09	F(1, 119)	=	261.56
Residual	2.2402e+09	119	18825282.1	Prob > F	=	0.0000
Total	7.1641e+09	120	59700977.7	R-squared	=	0.6873
				Adj R-squared	=	0.6847
				Root MSE	=	4338.8

price	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
inches	526.8323	32.57528	16.17	0.000	462.33	591.3346
_cons	-14299.58	1468.823	-9.74	0.000	-17208	-11391.17

ta-output 6.5 Regression of TV price in NOK (price) on TV size in inches (inches).

```
. gen log_price = ln(price)
```

```
. reg log_price inches
```

Source	SS	df	MS	Number of obs	=	121
Model	58.6466019	1	58.6466019	F(1, 119)	=	539.23
Residual	12.942466	119	.108760218	Prob > F	=	0.0000
				R-squared	=	0.8192
				Adj R-squared	=	0.8177
Total	71.5890679	120	.596575566	Root MSE	=	.32979

log_price	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
inches	.0574962	.002476	23.22	0.000	.0525934	.0623989
_cons	6.256483	.1116436	56.04	0.000	6.035417	6.477548

ta-output 6.6 Regression of log TV price (log_price) on TV size in inches (inches).

```
. reg log_price inches smart
```

Source	SS	df	MS	Number of obs	=	121
Model	64.2707866	2	32.1353933	F(2, 118)	=	518.15
Residual	7.31828133	118	.062019333	Prob > F	=	0.0000
				R-squared	=	0.8978
				Adj R-squared	=	0.8960
Total	71.5890679	120	.596575566	Root MSE	=	.24904

log_price	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
inches	.0480655	.0021158	22.72	0.000	.0438756	.0522554
smart	.4940654	.0518822	9.52	0.000	.3913245	.5968062
_cons	6.38027	.085303	74.80	0.000	6.211347	6.549193

ta-output 6.7 Regression of log TV price (log_price) on TV size in inches (inches) and the dummy smart TV (smart).

```
. use "C:\Users\700245\Desktop\temp_myd\fem_stud_sport.dta"
```

```
. reg phys_act_h par_exe_int
```

Source	SS	df	MS	Number of obs	=	293
Model	122.394756	1	122.394756	F(1, 291)	=	11.54
Residual	3085.52248	291	10.60317	Prob > F	=	0.0008
				R-squared	=	0.0382
				Adj R-squared	=	0.0348
Total	3207.91724	292	10.9860179	Root MSE	=	3.2563

phys_act_h	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
par_exe_int	.4652037	.136924	3.40	0.001	.1957168	.7346907
_cons	1.505246	.4920277	3.06	0.002	.5368623	2.47363

```
. reg phys_act_h par_exe_int exercise_int
```

Source	SS	df	MS	Number of obs	=	293
Model	960.895903	2	480.447952	F(2, 290)	=	62.01
Residual	2247.02133	290	7.74834942	Prob > F	=	0.0000
				R-squared	=	0.2995
				Adj R-squared	=	0.2947
Total	3207.91724	292	10.9860179	Root MSE	=	2.7836

phys_act_h	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
par_exe_int	.2007574	.1197773	1.68	0.095	-.0349856	.4365005
exercise_int	1.32725	.1275867	10.40	0.000	1.076137	1.578363
_cons	-3.634039	.6488283	-5.60	0.000	-4.911048	-2.357029

```
. reg exercise_int par_exe_int
```

Source	SS	df	MS	Number of obs	=	293
Model	22.4515335	1	22.4515335	F(1, 291)	=	13.73
Residual	475.990446	291	1.635706	Prob > F	=	0.0003
				R-squared	=	0.0450
				Adj R-squared	=	0.0418
Total	498.44198	292	1.70699308	Root MSE	=	1.2789

exercise_int	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
par_exe_int	.1992438	.0537792	3.70	0.000	.0933983	.3050893
_cons	3.872131	.1932521	20.04	0.000	3.491782	4.25248

ta-output 6.8 Three sets of regressions to estimate the path model in [Figure 6.10](#).

Notes

- 1 In real-life academic settings (yes, there is such a thing!), several controls would be included in this regression. I omit these here, and in the remaining chapter examples, to highlight the procedures of interest. The only complexity introduced by adding controls is that the regression coefficient of main interest might change in magnitude as we saw it did in [Chapter 3](#).
- 2 This procedure is the general one that works for all statistics programs, with small variations. Stata also has a built-in command that does these analyses in one step, as in:

```
"reg phys_act_h i.gender##c.friend_phys".
```

That is, the “##”-command tells Stata to multiply the two variables and add the resulting interaction term to the regression.

- 3 Since men are coded 0 in the data, the value of the product term variable for them also becomes 0. (Multiplying a number with 0 always yields 0.) Since women are coded 1 in the data, the value of the product term variable for them becomes -0.79 .

- 4 The command for generating the graph is:

```
reg phys_act_h i.gender##c.friend_phys
  margins gender, at(friend_phys=(0 4))
  marginsplot, noci title(Students' hours of exercise per week) xtitle(Friends' exercise level level (0 = lowest; 4 = highest)) ytitle(Hours per week)
```

- 5 A regression may contain several interaction variables (product terms), something which makes interpretation harder (but see Mitchell, 2012). Remember always to keep the original variables in the regression, i.e. x_1 and x_2 and $x_1 * x_2$. (Remember also that $*$ is the Stata-convention for the multiply sign, $\times$.)
- 6 Interaction terms may also be created between two dummies or two numerical variables, and they are roughly interpreted along the lines of the one presented here (for more on this, see Mitchell, 2012). The general idea is that the effect of x_1 on y is *conditional* on the level of another independent variable, x_2 . Interaction models are also called multiplicative models in the literature.
- 7 I once more omit x_2 and x_3 and so on for the sake of exposition, but it goes without saying that non-linearity issues could (and perhaps often should) be examined for every independent variable in the regression model, except for dummies.
- 8 The command for generating the graph is:

```
gen earnthousands = earnings_E/1000
  graph twoway (scatter earnthousands age) (lfit earnthousands age) (qfit earnthousands age), xtitle("Age of player") ytitle("Earnings, thousands of Euro")
```

- 9 There are more complex non-linear relationships, such as a cubic one. This requires the use of x , x^2 and x^3 (i.e. $x * x * x$) in the regression. But in practice the quadratic specification is by far the most commonly applied. Related but more complicated procedures involve piecewise regression models and spline models; see Lohmann (2015) for this.
- 10 The commands are:

```
gen age21_23 = age
gen age24_26 = age
gen age27_29 = age
gen age30_32 = age
gen age33ormore = age
recode age21_23 21/23 = 1 else = 0
recode age24_26 24/26 = 1 else = 0
recode age27_29 27/29 = 1 else = 0
recode age30_32 30/32 = 1 else = 0
recode age33ormore 33/39 = 1 else = 0
```

- 11 Again, x_2 and x_3 and so on are omitted for the sake of exposition.
- 12 A user-friendly introduction to logarithms for non-mathematicians is given by Studenmund (2017).
- 13 Logging y to get a more linear relationship between x_1 and y quite often also solves another issue. Not shown here, the linear regression in Stata-output 6.5 has severe heteroscedasticity problems. In contrast, the logarithmic regression in Stata-output 6.6 has no such problems. See King and Roberts (2015) for more on this.
- 14 A note on terminology: Sometimes the linear or plain vanilla model is called the lin-lin model, whereas the semi-logarithmic model is called log-lin. The double-logarithmic one is thus called log-log. There is also a fourth and less used semi-logarithmic variant, i.e. lin-log. The latter is also called a production function in economics (see

Studenmund, 2017). As always, theory, prior research and common sense (as well as the idiosyncrasies of the data in question) guide us regarding which of these functional forms (i.e. type of relationships) to prefer.

- 15 Dummy independent variables may be logged if they first are coded as 2.718228 and 1. Logging these values then yields 1 and 0.
- 16 More extensive models include control variables, such as age and gender. They could also include more variables similar to students' exercise interest, like attitudes towards dietary habits or smoking behavior.
- 17 Whether or not the indirect effect is statistically significant may be of interest. This requires hard calculations by hand, but fortunately Stata has an add-on that may be downloaded from the Internet (write "findit medeff" in the command window of Stata, press Enter and follow the instructions). The command (see the help file for details; "help medeff") and results appear below. We note that the indirect effect, called "ACME", has a 95% confidence interval ranging from about 0.12 to 0.44, making it (indirectly) statistically significant.
- 18 A more precise, formal name is an endogenous independent variable (more on this in [Chapter 9](#)).